Conversational Model for Indian Sign Language (ISL)

Industry / Inhouse Project : 2025-26

Review - I

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Group Members : 4

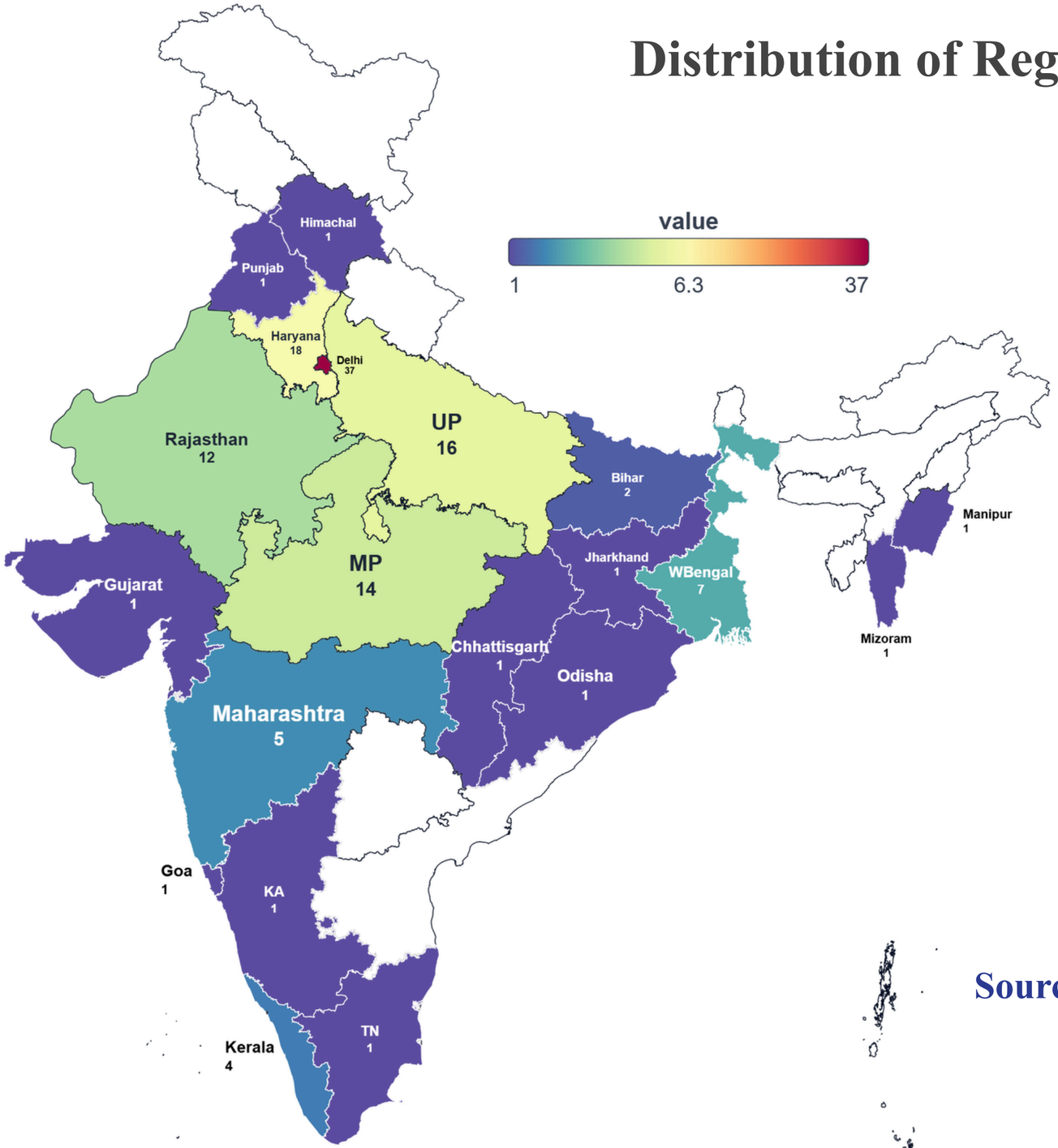
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Introduction to Project

- India has approximately **63 million** people suffering from significant auditory impairment, representing about 6.3% of the population.
- India has approximately **300 certified ISL interpreters**.
- Rapid advancements in Computer Vision, Deep Learning, and real-time gesture recognition enable automated interpretation of visual hand movements.
- Project Aim: Develop a **Conversational ISL Recognition Model** that converts ISL gestures into text and speech outputs to enable accessible and inclusive communication.

Distribution of Registered ISL Interpreters in India



Key Insights :

- **ISL interpreters are unevenly distributed across states**
- **Northern cluster shows relatively higher availability**
- **Majority of states have minimal interpreter presence**

Source: ISLRTC Directory of Registered ISL Interpreters

Lacuna in the existing system

Despite the availability of some assistive technologies, several limitations still exist in current communication systems for hearing-impaired individuals:

- Limited availability of trained sign language interpreters
- Existing solutions are often expensive and not easily accessible
- Many systems lack real-time gesture recognition
- Poor portability and dependency on manual intervention
- Limited scalability and lack of user-friendly interfaces

These gaps result in communication delays, reduced independence, and social exclusion for hearing-impaired individuals.

Problem Definition

- Hearing-impaired individuals face difficulties communicating with people who do not understand Indian Sign Language (ISL).
- Shortage of interpreters increases misunderstandings and communication barriers.
- This leads to limited opportunities and reduced quality of life.
- Existing solutions are not fully cost-effective, portable, or real-time.

Therefore, there is a need for a scalable and real-time gesture recognition system that can convert ISL gestures into text and speech to promote inclusive communication.

Literature Survey

Indian Sign Language (ISL)

1.Detection (Hand & Gesture Detection)

Paper / Study	Dataset Used	Model	Parameters	Evaluation Metrics	Result	Limitations
ISL Hand Detection using YOLOv5	ISL Alphabet Dataset, Custom ISL Videos	YOLOv5	Learning rate, batch size, confidence threshold, IoU	Precision, Recall, mAP, FPS	mAP > **90%, ~25**–30 FPS (real-time)	Poor performance in low lighting & occlusion
Real-time Hand Detection using MediaPipe	Custom ISL Dataset	MediaPipe Hands	21 keypoints, frame rate tuning	Precision, FPS	**~30** FPS, lightweight & fast	Not robust in complex backgrounds
ISL Detection using Faster R-CNN	Kaggle ISL Hand Gesture Dataset	Faster R-CNN	Anchor size, learning rate	Precision, Recall, mAP	High detection accuracy (**~88–92%**)	Computationally expensive

2.Recognition (Static & Dynamic Gestures)

Paper / Study	Dataset Used	Model	Parameters	Evaluation Metrics	Result	Limitations
ISL Alphabet Recognition using CNN	ISL Alphabet Dataset	CNN	Epochs, filters, kernel size, dropout	Accuracy, Precision, Recall, F1-score	Accuracy ~**95%**	Works only for static gestures
Dynamic ISL Recognition using CNN+LSTM	Self-recorded ISL Dataset	CNN + LSTM	Sequence length, hidden units	Accuracy, F1-score	~**92%** accuracy	Needs large labeled video data
ISL Recognition using 3D-CNN	ISL CSL Dataset	3D-CNN	Temporal window size, stride	Accuracy	~**94%** accuracy	High computational cost

3. Translation (Sign → Text)

Paper / Study	Dataset Used	Model	Parameters	Evaluation Metrics	Result	Limitations
ISL Sentence Translation using BiLSTM	ISL Sentence Dataset	CNN + BiLSTM	Hidden units, embedding size	WER, Accuracy	WER ~**18%**	Poor long sentence translation
Transformer-based ISL Translation	Word-level ISL Dataset	Transformer (Seq2Seq + Attention)	Layers, attention heads	BLEU, WER	BLEU ~**30–35**	Requires large annotated dataset

4. Interpretation (Sign → Text → Speech)

Paper / Study	Dataset Used	Model	Parameters	Evaluation Metrics	Result	Limitations
ISL Interpretation System	ISL Video + Text Corpus	CNN + LSTM + NLP + TTS	Speech rate, decoding length	Accuracy, BLEU, Latency, User Satisfaction	Real-time prototype (**~1–2** sec latency)	Limited vocabulary

Methodology Employed

- **Dataset Collected:** ISign Dataset.
- **Data Preprocessing:** Perform data cleaning, labeling, resizing, and normalization to improve model performance.
- **Model Development:** Implement deep learning algorithms such as Convolutional Neural Networks (CNN) or MediaPipe-based hand tracking for accurate gesture detection.
- **Training and Validation:** Train the model using prepared datasets and evaluate it using performance metrics like accuracy and precision.
- **Gesture Translation Module:** Convert recognized gestures into readable text and generate speech output using text-to-speech technology.
- **System Testing:** Test the system under varying lighting conditions, backgrounds, and user scenarios to ensure reliability.
- **Deployment:** Develop a mobile or web-based interface to provide a user-friendly experience.

Hardware, Software, Tools and constraint

- **Hardware:**

  Laptop/PC, Webcam/Mobile Camera, GPU

- **Software:**

  Python, OpenCV, MediaPipe, TensorFlow, PyTorch, Android Studio, NLP Libraries

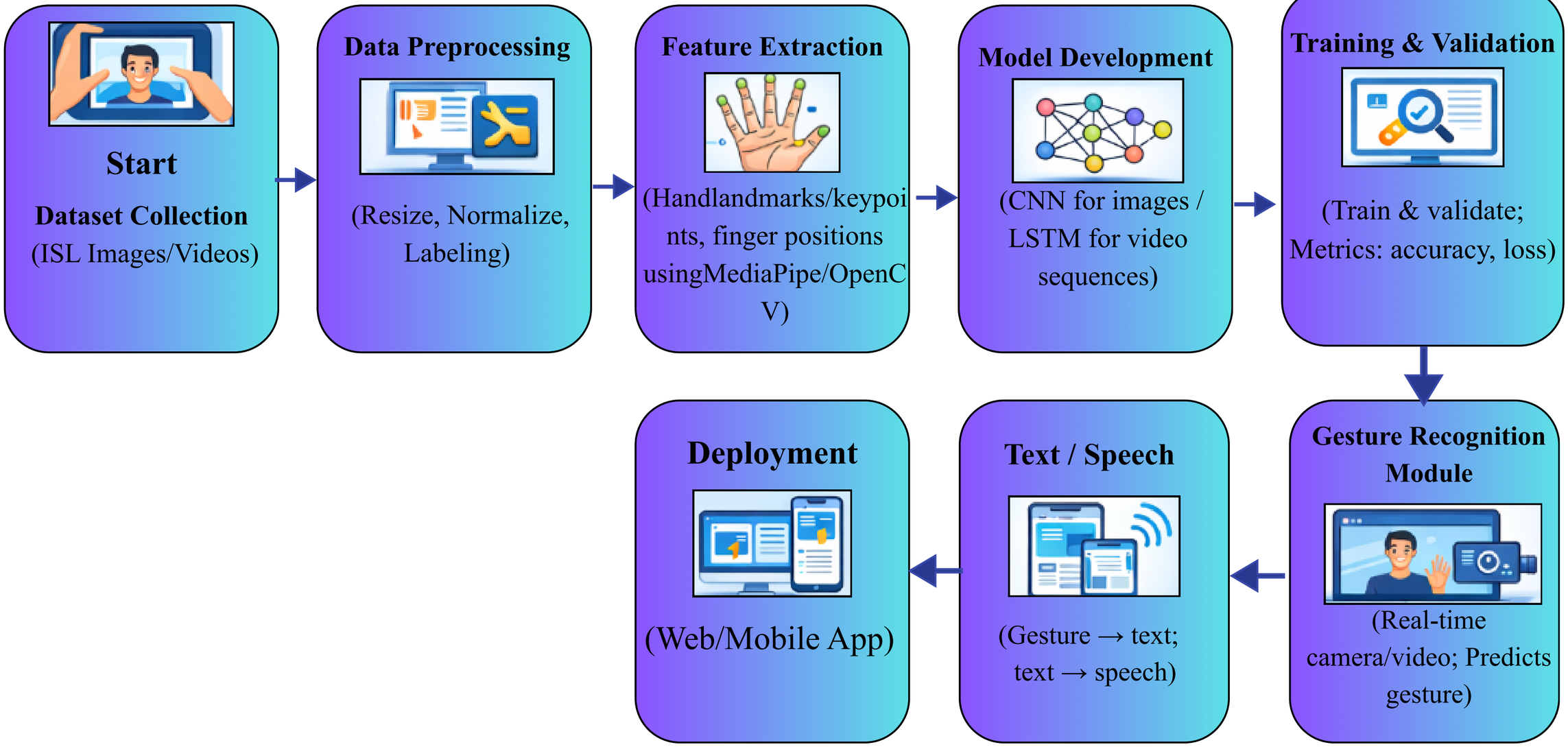
- **Tools:**

  VS Code, Jupyter Notebook/Google Colab, Git & GitHub, Postman

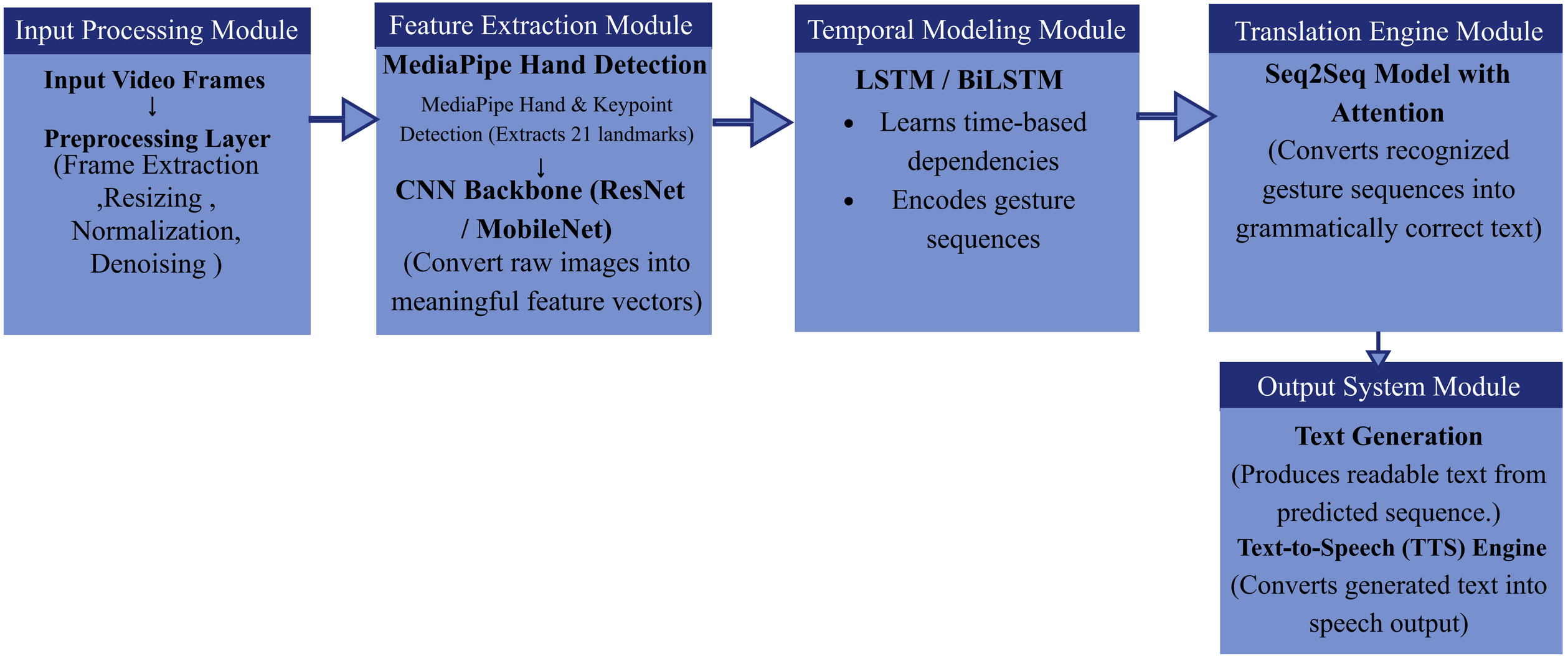
- **Constraints:**

  Limited dataset size, Controlled testing environment, Hardware limitations, Real-time accuracy challenges, Academic time constraints

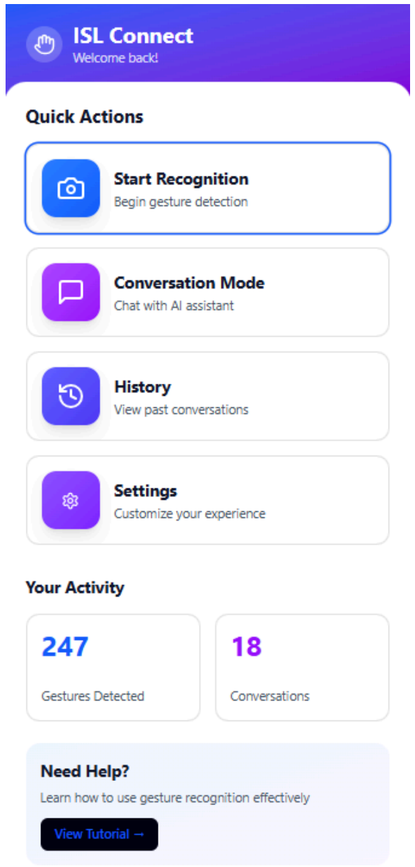
Block Diagram

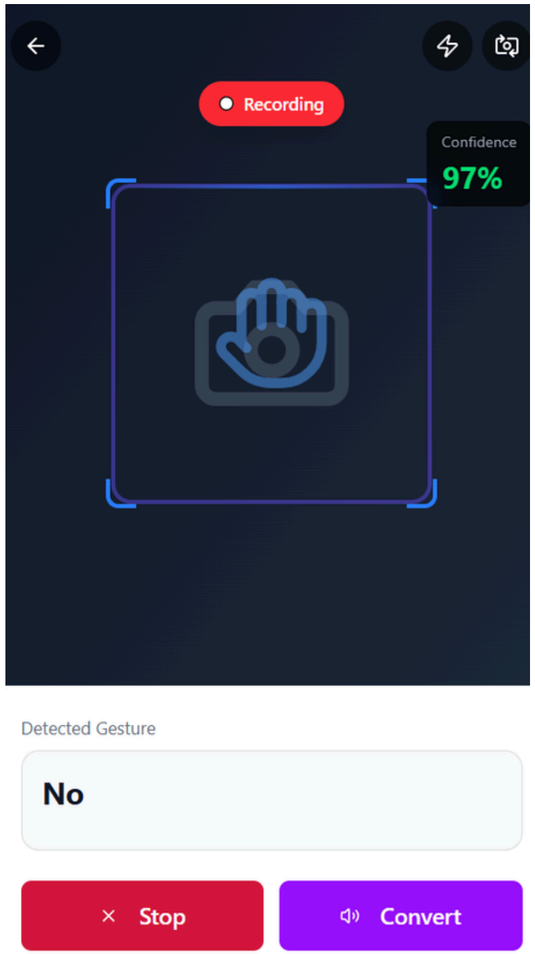


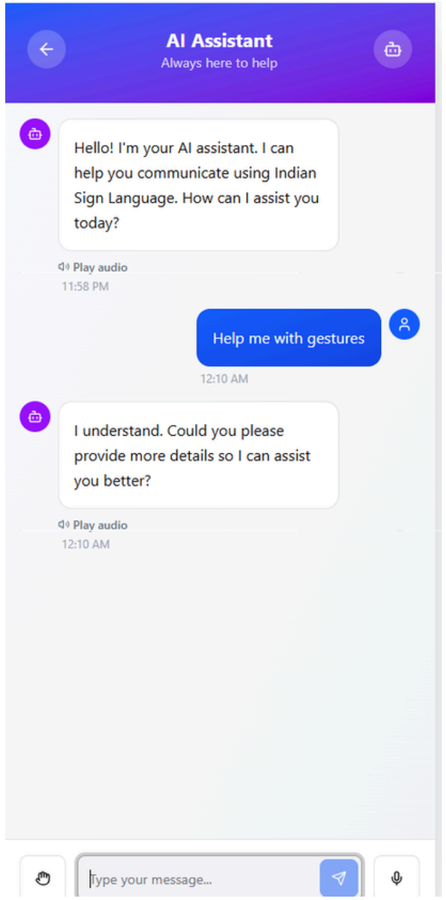
Modular Diagram



FIGMA / UI



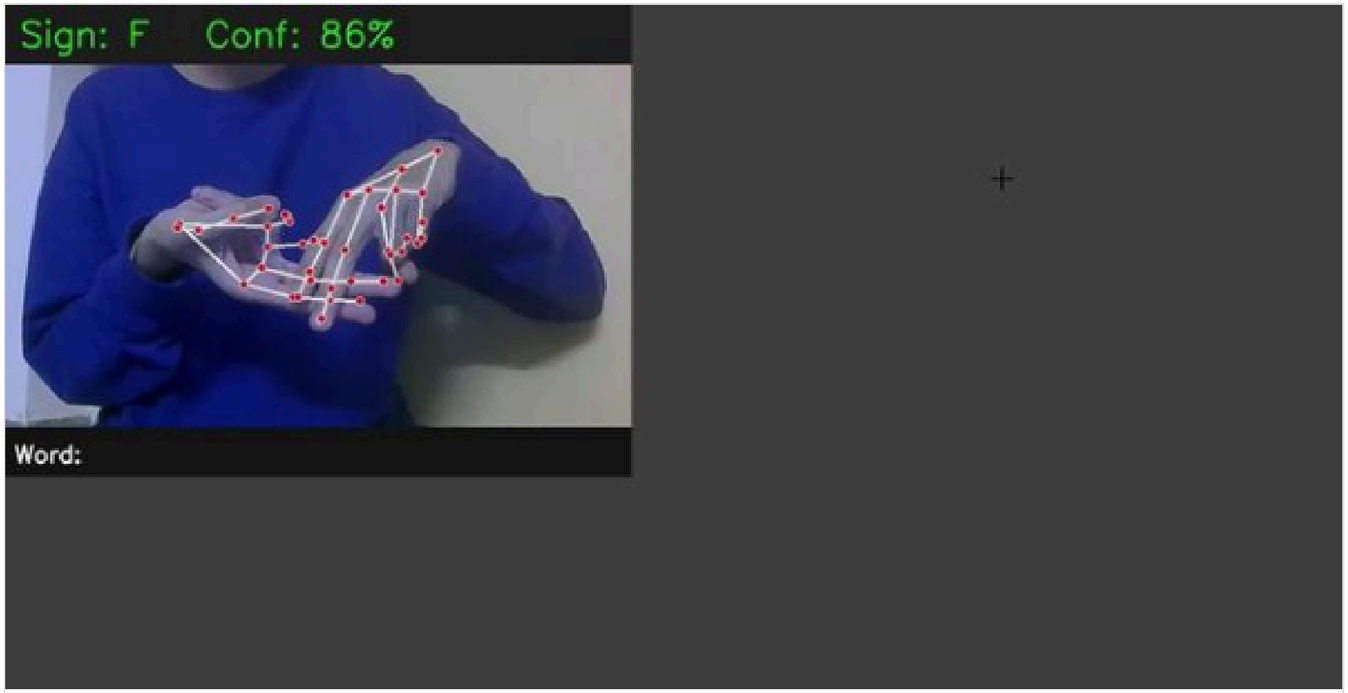




Figma Link

Work Done

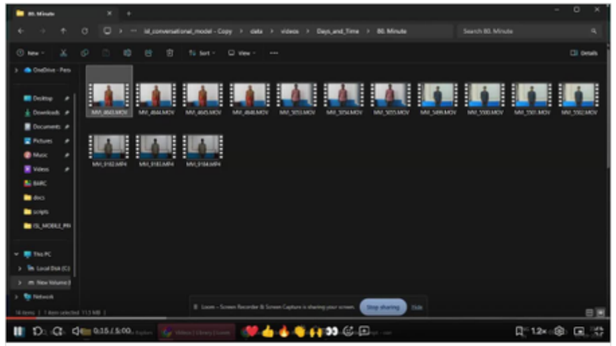
A CNN-based classifier was designed and trained on the preprocessed landmark data.



Adobe Express - ISL Real-Time Recognition 2026-03-24 23-21-17.mp4

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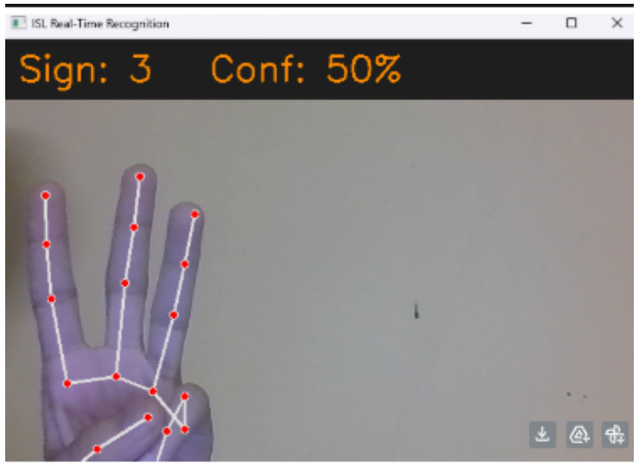
Dropbox



Screen Recording 2026-09-02 221331.mp4

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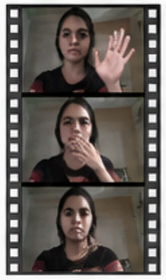


ISL RECOGNITION MODEL ARCHITECTURE (V2)

From Video → Body-Centered Landmarks → Spatiotemporal Features → 22-Class Prediction

1. INPUT: VIDEO SEQUENCE

Variable length video recorded from webcam or dataset



Variable # Frames
(≈ 2 – 4 sec)

2. MEDIAPIPE HOLISTIC

Extract 3D landmarks for pose, face & hands



- 33 Pose Landmarks
- 21 Left Hand Landmarks
- 21 Right Hand Landmarks

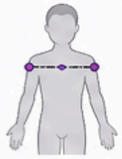
(+ Face landmarks used internally by Mediapipe)

3. BODY-CENTERED & SCALE NORMALIZED FEATURES (V2)

Compute Body Center (Shoulder Midpoint)



Compute Body Scale (Shoulder Distance)



Per Frame: 386 Features

Pose	Left Hand	Right Hand	Left Hand	Right Hand
Body-Relative	Body-Relative	Body-Relative	Wrist-Relative	Wrist-Relative
33 x 4 = 132	21 x 3 = 63	21 x 3 = 63	21 x 3 = 63	21 x 3 = 63

Hand Presence Flags (2)
[Left Present, Right Present]

4. TEMPORAL RESAMPLING

Resample / Interpolate to fixed length

Frame 1 (386)
Frame 2 (386)
⋮
Frame T original (386)

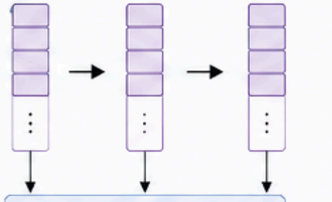
→

Fixed Length T = 40 Frames

Frame 1 (386)
Frame 2 (386)
⋮
Frame 40 (386)

5. TEMPORAL ENHANCEMENT

Add velocity and acceleration (First & Second derivatives)



Concatenate on Feature Axis

Enhanced Feature Per Frame
386 x 3 = 1158

6. SPATIOTEMPORAL MODEL (TCN + BiGRU + ATTENTION POOLING)

Input: (40, 1158)
(40 frames, 1158 features)

Temporal Convolution Block 1 (Dilated Conv1D)
Filters=64, Kernel=3, Dilation=1, ReLU + BN + Dropout

Temporal Convolution Block 2 (Dilated Conv1D)
Filters=128, Kernel=3, Dilation=2, ReLU + BN + Dropout

Temporal Convolution Block 3 (Dilated Conv1D)
Filters=256, Kernel=3, Dilation=4, ReLU + BN + Dropout

TCN captures local temporal patterns at multiple time scales (short → long range).

Bidirectional GRU (Return Sequences)
Units=256, Dropout

Bidirectional GRU (Return Last)
Units=256, Dropout

BiGRU captures forward and backward temporal dependencies and overall sequence context.

Attention Pooling
(Learn important time steps)

Attention Pooling learns which time steps are most informative for the prediction.

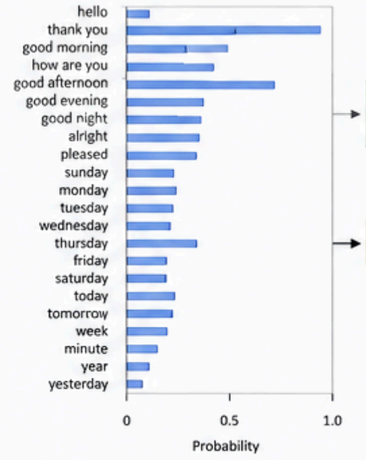
Dense (256) + ReLU + Dropout

Dense (22) + Softmax
22 ISL Classes

Dense layers learn high level representation and map to 22 class probabilities.

7. OUTPUT

Probability over 22 Classes



Predicted Class
(Highest Probability)

Confidence Score
(%)

TRAINING DETAILS

- **Dataset** INCLUDE Dataset (22 classes, 11,564+ videos)
- **Input Shape** : (40, 1158)
- **Loss Function** : Categorical Crossentropy
- **Optimizer** : Adam (lr=1e-4)
- **Batch Size** : 32
- **Epochs** : 50 (Early Stopping)
- **Cross Validation** : 5-Fold
- **Test Accuracy (Held-out)** 92.59%

INFERENCE (WEBCAM / REAL-TIME)

1. Capture frames from webcam.
2. Extract landmarks using Mediapipe Holistic.
3. Compute body-centered normalized 386 features per frame.
4. Resample to 40 frames.
5. Compute velocity & acceleration → 1158 features.
6. Normalize using training mean & std.
7. Run TFLite model → Get class probabilities.
8. Apply confidence threshold and smoothing if needed.

WHY THIS ARCHITECTURE WORKS WELL?

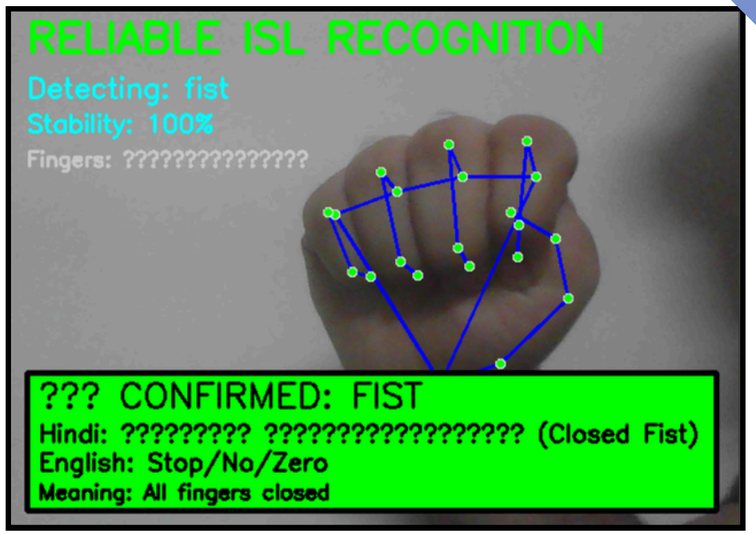
- ✅ Body-centered normalization → Invariance to position, scale & distance.
- ✅ Velocity & acceleration → Captures motion & dynamics of signs.
- ✅ TCN → Multi-scale temporal pattern learning.
- ✅ BiGRU → Long-range dependencies in both directions.
- ✅ **Attention Pooling** → Focus on the most informative frames.
- ✅ **Compact & efficient** → Real-time inference on CPU.

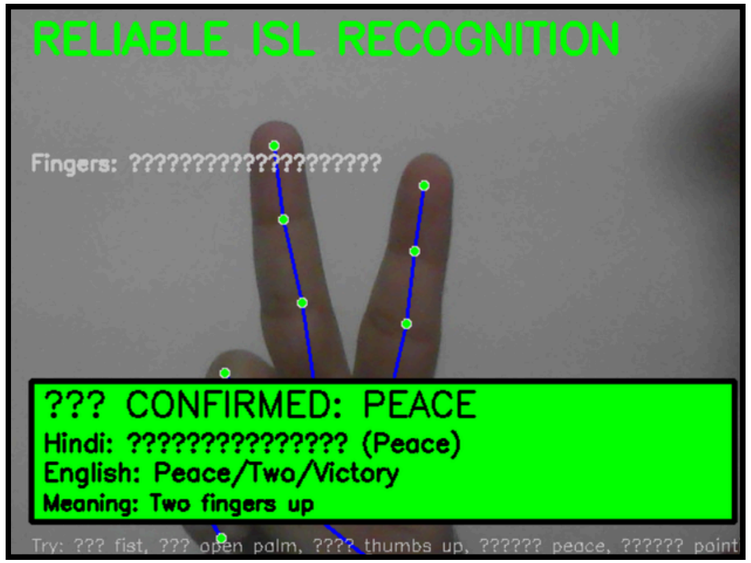
Legend:

Input / Preprocessing · Feature Construction · Model Layers · Output / Decision · Training / Inference Info

Next Work Plan

- **Expand Gesture Vocabulary:** Increase the dataset to train the model on a wider range of alphabets, numbers, and common everyday ISL words.
- **Enhanced Recognition Accuracy**: Optimize model parameters and expand training data to ensure high precision and reliability across a wider range of users and environments.
- **Live Web/Mobile Integration:** Develop a user-friendly application interface that seamlessly accesses device cameras for on-the-go, real-time translation.
- **Sentence-Level Conversation Support:**Implement word sequencing for complete sentence formation and Improve context understanding for natural conversation flow.





Conclusion

The proposed system aims to **reduce** the communication gap between **hearing-impaired and hearing individuals** by providing real-time sign language recognition and translation. By converting **gestures into** understandable **text and speech**, the system enables **smoother** and more **effective** everyday communication.

The solution emphasizes **accessibility, reliability**, and **user-friendly** interaction, making it suitable for practical **real-world** use. With scalable features and conversational support, it has the potential to enhance independence and confidence among hearing-impaired users.

Overall, this project represents a meaningful step toward creating a more **inclusive** and **barrier-free communication** environment.

References

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[2] Anand, Nisha. *A Sociolinguistic study of the use of Indian Sign Language*. Diss. M. Phil Thesis. Jawaharlal Nehru University. New Delhi, 2016.

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[7] Official ISL website : https://indiansignlanguage.org/

Thankyou !